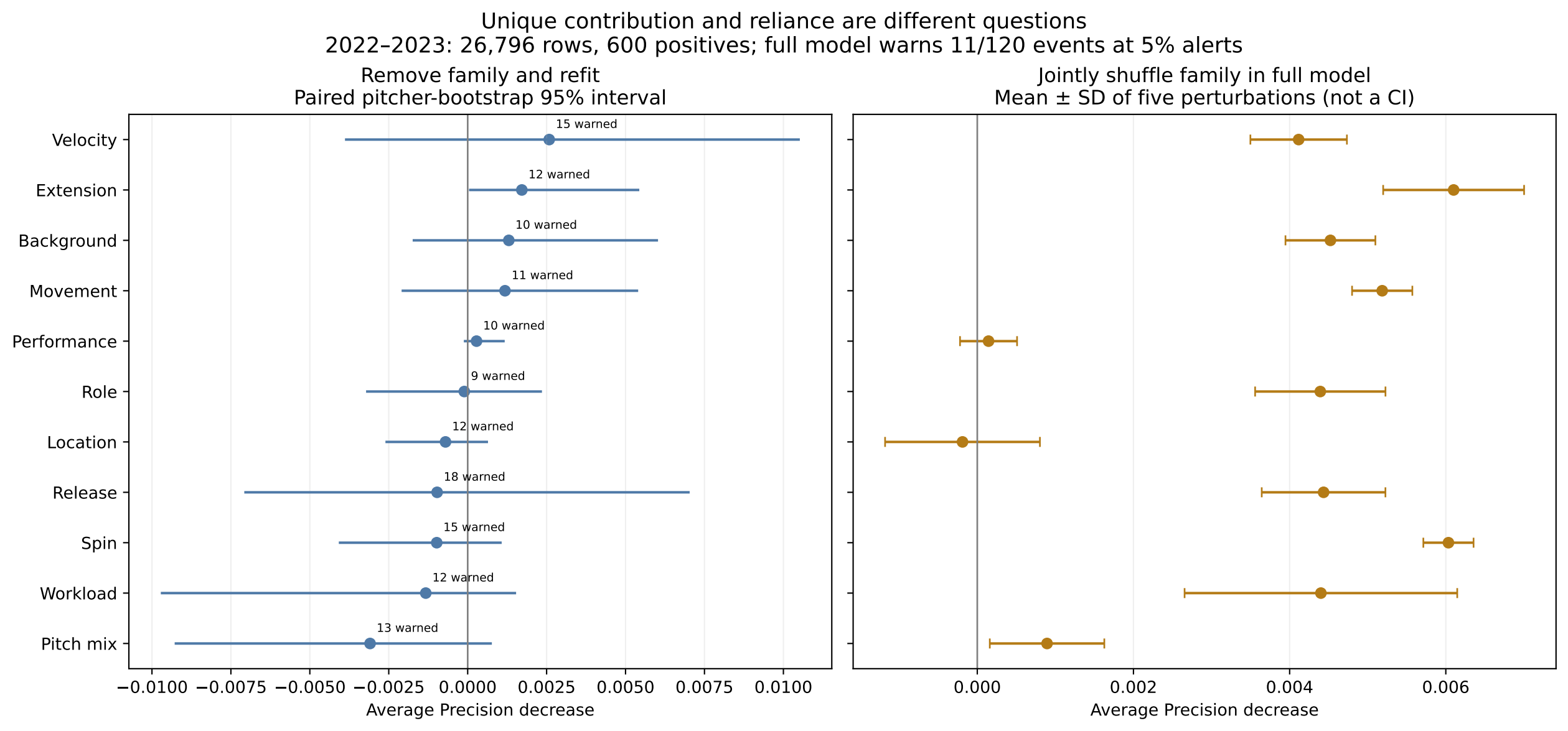
Unique contribution and reliance are different questions
2022–2023: 26,796 rows, 600 positives; full model warns 11/120 events at 5% alerts
Remove family and refit
Paired pitcher-bootstrap 95% interval
Jointly shuffle family in full model
Mean ± SD of five perturbations (not a CI)
Velocity
Extension
Background
Movement
Performance
Role
Location
Release
Spin
Workload
Pitch mix
15 warned
12 warned
10 warned
11 warned
10 warned
9 warned
12 warned
18 warned
15 warned
12 warned
13 warned
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
Average Precision decrease
0.000 0.002 0.004 0.006
Average Precision decrease